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The genome of the vining fern *Lygodium microphyllum* highlights genomic and functional differences between life phases of an invasive plant

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Functional and genomic studies on the differences between the gametophyte and sporophyte life phases of plants remain scarce, yet unraveling these dynamics is crucial to understanding the biology of plants and the success of each phase under different conditions. Here, we provide a reference genome for the highly invasive fern *Lygodium microphyllum* and compare the transcriptomic and epigenomic landscapes of its gametophyte and sporophyte life phases. Under ambient conditions, we found differential regulation and splicing of developmental genes (homeobox and MADS-box clades) may play a role in the genomic determination of the haploid and diploid life stages. We generated a base pair-resolution methylome of a fern gametophyte, and determined that methylation patterns are remarkably similar between vegetative tissues despite their morphological and functional differences. We further explored the physiological and transcriptomic responses of gametophytes and sporophytes to freezing stress, the most likely abiotic factor limiting further range expansion of this invasive species. While controlled by the same genome, we show that life phases and tissues use alternative molecular pathways in response to freezing and greater physiological resilience in the gametophyte life stage to this stressor. Our results underscore the need to incorporate both life phases when developing effective mitigation strategies, as differential responses to environmental stressors between phases reveal opportunities for management approaches (e.g., targeting gametophytes in addition to sporophytes). These genomic resources fill a gap in our understanding of fundamental plant biology and inform invasive species research.

alternation of generations | fern | genome | epigenomics | invasive species

Significance

All land plants undergo an alternation of generations between haploid gametophyte and diploid sporophyte life phases. How these disparate life phases are generated from a single genome, and the functional implications of these differences for plant success, is largely unknown. Moreover, understanding life-phase-specific differences in rapidly evolving populations, such as invasive species, is critical to developing effective management strategies. We assembled a chromosome-level genome and generated epigenomic and transcriptomic resources from gametophyte and sporophyte phases of the invasive fern *Lygodium microphyllum,* which is estimated to cost the United States more than $2 million annually. We highlight transcriptomic, epigenomic, and physiological variation between life phases and find support for the crucial role of the often overlooked gametophyte in the invasion process.

All land plants alternate between multicellular diploid sporophyte and haploid gametophyte life phases (typically referred to as the "alternation of generations" life cycle). Variations in this life cycle mark pivotal transitions in the evolution of plants (e.g., the transition from gametophyte-dominance in bryophytes to sporophyte-dominance in vascular plants), but very little is known about how a single underlying genome generates these radically disparate stages. Ferns are an excellent system for unraveling these processes as they have free-living, nutritionally independent gametophytes and sporophytes (1). Previously, researchers have found minimal but distinct differences in gene expression levels between fern life stages [e.g., up to 90% of genes had no significant differences in expression level (2–4)], but the regulation and functional roles of these genes remains a crucial knowledge gap. This is in part because genomic resources for this clade still lag behind other major lineages despite recent publications of high-quality fern genomes (3, 5–8).

The gametophyte has long been thought of as a fragile handicap in the fern life cycle (9, 10), and has largely been overlooked by researchers. The gametophyte generation, however, is the site of sexual reproduction and is thus pivotal for the establishment of sporophyte populations and will shape evolutionary trajectories (11). Furthermore, physiological evidence suggests that the haploid life phase of ferns may in fact be more hardy than the sporophyte, able to withstand extreme desiccation and drought (12, 13), freezing (14, 15), and heat (13). These differences highlight a largely neglected fundamental aspect of organismal plant biology, with profound functional implications for our understanding of the processes that shape the niche, distribution, and evolution of ferns.

The application of high-throughput sequencing to explore the genomic underpinnings of these functional differences is underexplored yet has provided valuable insights in other groups, including improving our understanding of the regulation of the alternation of generations in algae (16), bryophytes (17, 18), and angiosperms (19), and has revealed large-scale genomic reorganization associated with transitions in the life cycles of *Arabidopsis* (20) and

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The authors declare no competing interest.

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animals (21). Of particular interest is how differences between life stages play out in rapidly evolving systems such as invasive species, which pose an unprecedented threat to ecological, environmental, and economic systems globally (22). Recent estimates place the economic costs of biological invasions at over $100 billion annually world-wide (23). Given the functional variation between life stages in many organisms, determining the relative roles of the gametophyte and sporophyte generations in the invasion process of ferns may have implications for invasion biology generally, including in the development of effective management strategies.

Here, we present a high-quality, chromosome-scale reference genome for the invasive fern *Lygodium microphyllum* (Fig. 1). A member of the order Schizaeales, *Lygodium* occupies a pivotal phylogenetic position as sister to the core leptosporangiate ferns, which comprise the vast majority of fern diversity (24) (and, to date, all published fern genomes are core leptosporangiates). Comparisons of lineages sister to major clades (e.g., *Amborella*) provide valuable opportunities to investigate ancestral genome organization and content. Therefore, investigations of the life stages of *L. microphyllum* offer insights into the differences between generations that predate most of extant fern diversity.

Lygodium microphyllum is native through much of the eastern tropics (eastern Asia, Australia, Africa) and was initially introduced to the United States as an ornamental plant, but quickly escaped cultivation and naturalized in Florida in the 1960s (25). The species has a vining, indeterminate growth form and extensive rhizomatous proliferation, which facilitate the formation of dense mats that smother and shade native ecosystems, disrupt natural

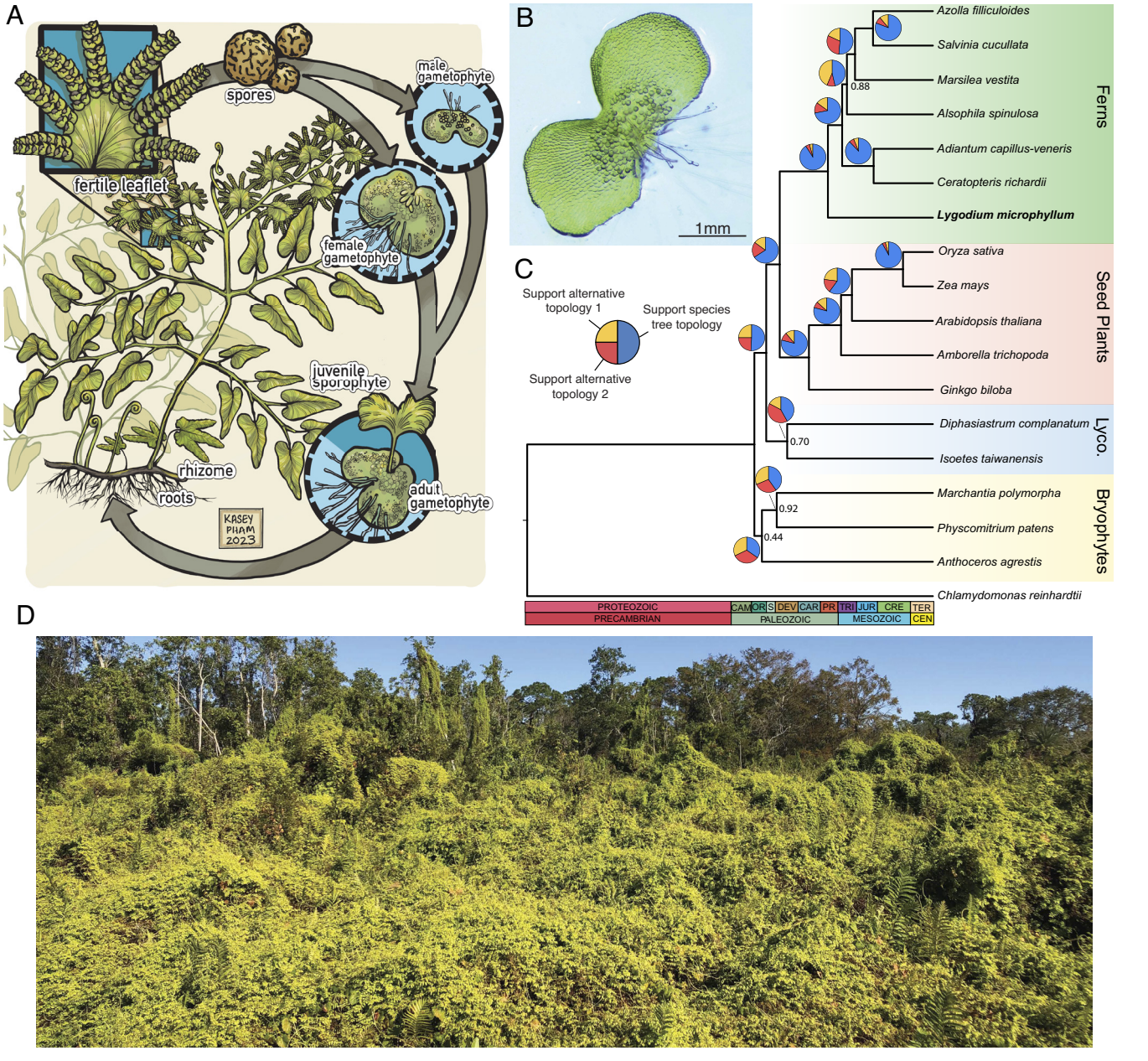


Fig. 1. Genome assembly of *L. microphyllum*. (*A*) The life cycle and unique morphology of *L. microphyllum* (illustration by Kasey Pham). The sporophyte-dominated life cycle alternates between the diploid sporophyte and haploid gametophyte phases. The vining habit is facilitated by dormant pinnae buds and circumnutation, depicted here at the apex of the leaf. (*B*) Antheridate gametophyte of *L. microphyllum*. (Scale bar, 1 mm.) (*C*) Phylogenetic placement of *L. microphyllum* within the Viridiplantae on a time-calibrated tree. Major lineages of land plants are delineated by background color. Pie charts at nodes show the proportion of gene trees that support each topology (species tree topology or the two alternative topologies) and node support are local posterior probabilities. (*D*) Field site heavily impacted by *L. microphyllum* invasion in Florida (the light green plant dominating the fore- and mid-ground).

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Table 1. Genome assembly and annotation statistics

Metric	Value
Final assembly size	4.75 Gb
Number of contigs	1,801
Contig N50	17.35 Mb
Number of scaffolds/Unplaced contigs	952
Number of chromosome-level scaffolds	30
Scaffold N50	155.01 Mb
Repetitive content	78.59%
Genome BUSCO (Viridiplantae ODB10, *n* = 425) [No. single complete, No. duplicated complete]	98.11% [409S, 8D]
Number of predicted protein-coding genes	22,827
Proteome BUSCO (Viridiplantae ODB10, *n* = 425) [No. Single Complete, No. Duplicated Complete]	98.80% [400S, 20D]
Proteome OMArk (Tracheophyta, *n* = 6,460)	91.97%

fire regimes, and alter native community composition through canopy collapse (26). In its invasion of the United States, *L. microphyllum* is currently restricted to southern Florida, and freezing is widely recognized as range-limiting for its invasive populations (27, 28). We developed genomic resources to explore differences in life stages at multiomic scales from epigenomes to transcriptomes, and highlight how different physiological and genetic responses of the gametophyte and sporophyte to cooling and freezing may have important implications for management in a changing climate.

Results and Discussion

Genome Assembly and Annotation. The haploid genome size of *L. microphyllum* was estimated to be between 4.2 and 5.1 Gb, with low heterozygosity (0.106%) based on *k*-mer approaches (*SI Appendix*, Fig. S1), which are similar to flow cytometry-based estimates of 5.56 Gb (29). Using a combination of error-corrected Oxford Nanopore (37× coverage) and Illumina (60× coverage) reads, we generated a contig-level assembly that was 4.75 Gb in total length with a contig N50 of 17.37 Mb (Table 1 and Dataset S1). By integrating chromatin conformation capture data (Dovetail Omni-C; 73× coverage), we scaffolded the genome to the chromosome scale (*SI Appendix*, Fig. S2), with a total of 4.61 Gb (96.94%) anchored on 30 pseudomolecules representing the 30 chromosomes in the *L. microphyllum* karyotype (30, 31). The final assembly contained 952 scaffolds with a scaffold N50 of 155.01 Mb, and a plastome assembly of 163 kb (*SI Appendix*, Fig. S3). Of the 30 chromosome-level scaffolds, we identified high proportions of telomeric repeats $(AAACCCT)_n$ on at least one end for 28 scaffolds, and on both ends of 11 scaffolds (*SI Appendix*, Fig. S4). The genome assembly was highly complete, with 98.11% single and duplicated intact BUSCO genes (Viridiplantae ODB10, *n* = 425). The mapping rate of genomic short-read Illumina data was over 99%, and a *k*-mer based assessment indicated a highly complete and accurate assembly (QV = 45.4; *k*-mer completeness = 95.25%).

As with most plants, the majority of the *Lygodium* genome was composed of repetitive elements: 78.59% of the genome was occupied by repeats. Most of these elements were classified as long terminal repeats (59.97% of the genome) in Gypsy (33.77%) and Copia (17.94%) families, with a substantial proportion of unclassified repeats (12.13%). A small percentage of repeats were DNA transposons (3.91% of the genome), largely classified as CMC-EnSpm elements (3.30%; see Dataset S2). On average, repeat density was high (1,092 repetitive elements/Mb), while gene density was low (4.89 genes/Mb). The structural gene annotation using nine tissue types (Dataset S3) identified 22,827 genes, which is slightly less than the number of genes annotated in other diploid ferns genomes (3, 5, 7). Nearly all genes (22,625, 99.11%) were located on the thirty chromosome-level scaffolds. The predicted proteome was highly complete, with 98.80% BUSCO and 91.97% OMArk scores (Table 1). Gene metrics (e.g., length, number of exons) are similar to those in other published fern genomes (Datasets S4–S7).

Several recent studies of transcriptomic datasets have identified a putative whole genome duplication (WGD) shared by nearly all leptosporangiate ferns likely dating to the Permian or Triassic (32–36). We do not find evidence of this paleopolyploidy event in the *Lygodium* genome, with a clear lack of conserved synteny and no burst of duplicates in the K_S plot (Fig. 2). How rediploidization occurs in ferns, however, is largely unknown and different lineages support a rediploidization continuum from remarkably well-conserved synteny in *Alsophila spinulosa* (6) to massive genomic reorganization in *Ceratopteris richardii* (7) and *Adiantum capillus-veneris* (3). It appears that there is a fundamental biological difference in this process between flowering plants and ferns, with some hypotheses suggesting that ferns primarily silence or pseudogenize duplicated genes but retain chromosomal material (37). Like *Adiantum* and *Ceratopteris,* the Schizaeales have comparatively high rates of molecular evolution (32), especially relative to tree ferns, possibly linked to short generation times and herbaceous growth habit. Interestingly, *L. microphyllum* also has the smallest genome size reported in *Lygodium* (38). Taken together, this putatively derived reduction in genome size and accelerated rate of molecular evolution may obscure signatures of paleopolyploidy. Further study on genic fractionation and chromosomal retention will be required to determine why ferns differ so dramatically from other plant lineages in the rediploidization process.

Transcriptomic Landscapes of Fern Life Phases. How a single genome can produce two vastly different life phases has long fascinated, and continues to challenge, plant biologists (39). The differential regulation of the plant developmental program is likely a central player, although the mechanism through which this happens has yet to be determined. To compare the transcriptomic landscape of the gametophyte and sporophyte life phases, we mapped RNAseq reads generated from sterile leaves (hereafter "sporophyte" in this section) and pooled immature and mature gametophytes grown under the same ambient conditions to the reference genome and compared their expression levels and splicing patterns.

A total of 1,893 (9.5%) genes had significantly different expression levels between life phases: 5.7% of all genes (1,136) were upregulated and 3.8% (757) were downregulated in the gametophyte relative to the sporophyte. Our findings are similar to those from previous transcriptome and genome-based studies of differential gene expression between life phases in ferns: A transcriptomic study of *Polypodium amorphum* (2) found 9.8% of genes to be differentially expressed, and genome-based studies of *A. capillus-veneris* (3) and *C. richardii* (7) found 12.6% and 5.8% of genes differentially expressed, respectively. We further explored whether genes that were differentially expressed in gametophyte

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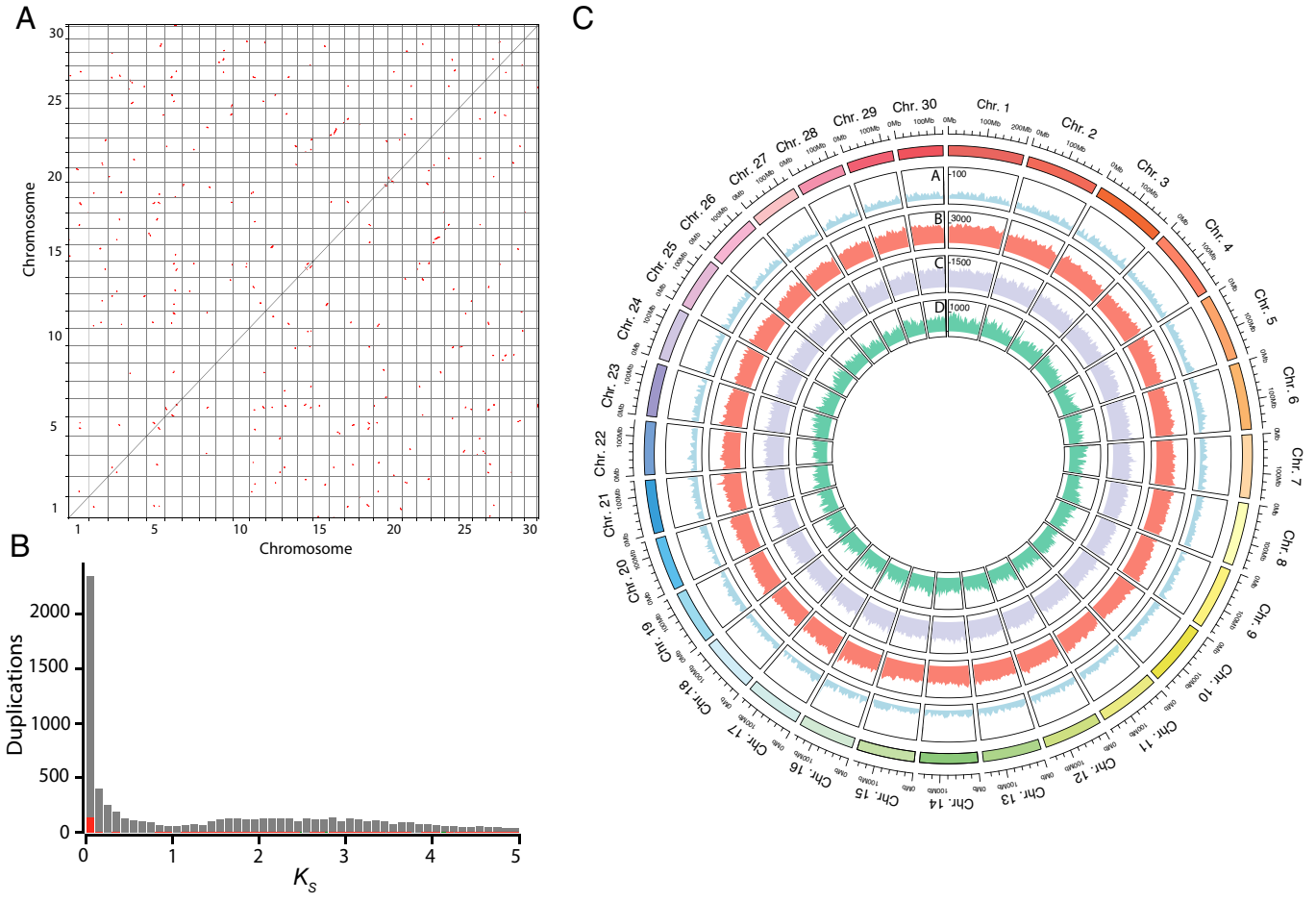


Fig. 2. No evidence of WGD from the *L. microphyllum* genome. (*A*) Dot plot of self–self synteny. (*B*) Plot of synonymous substitution rates (K_S) for all duplicated genes (gray) and syntenic duplicates (red). Analyses of synteny (*A*) and K_S(*B*) do not support a recent WGD in *L. microphyllum*. (*C*) Circos plot of the genome assembly of *L. microphyllum,* showing attributes of the 30 chromosome-level scaffolds. From the outside in, tracks are (*A*) gene density, (*B*) total repeat density, (*C*) Gypsy family repeat density, and (*D*) Copia family repeat density. Densities were calculated using a 5 Mb window size.

and sporophyte tissues were similar across the *Lygodium, Ceratopteris,* and *Adiantum* genomes. Using a phylogenomic approach, we identified 81 orthogroups that contained genes that were downregulated in the sporophyte relative to the gametophyte across the genomes of *Lygodium*, *Ceratopteris*, and *Adiantum*, and 520 orthogroups were shared by two of these three species (*SI Appendix*, Fig. S5 and Dataset S8). In comparison, 26 orthogroups contained genes that were upregulated in all three species, and 261 were shared by two of the three species (*SI Appendix*, Fig. S5 and Dataset S9). One of these orthogroups, annotated as the gibberellic acid-stimulated *Arabidopsis* (GASA) gene family, contained both consistently up- and down-regulated genes across all three genomes. Among the orthogroups that were consistently upregulated in the leaf relative to the gametophyte were annotated as gene families involved in stomatal development (*SCREAM2*), stomatal movement (*EULS3*), and organ patterning (*KNOX I/II*). In contrast, orthogroups that were consistently downregulated in the leaf relative to the gametophyte were annotated as gene families involved in embryo development (*PEL3, FLA7, LEA*), repression of *KNOX* expression (*OFP*), response to and biosynthesis of hormones (*GASA, ABC33, HAK1*), and growth (*EXO, THESEUS*). In total, 105 orthogroups contained genes that were consistently significantly differentially expressed, and, given the phylogenetic position of *Lygodium* sister to the core leptosporangiates, suggests that these expression levels may have been conserved over evolutionary time.

Within *Lygodium,* most pathways that were overrepresented in differentially expressed genes were related to metabolism (Datasets S10 and S11). Of particular interest, the diterpenoid biosynthesis pathway was enriched in upregulated genes in the gametophyte (8.61-fold enrichment [F.E.], adjusted P = 0.0019; *SI Appendix*, Fig. S6), which is involved in the generation of gibberellins. Gibberellins (GAs) are regulators of plant growth and development (40), and have been implicated in antheridiogen production in fern gametophytes (41, 42). The expression of genes such as mono- and dioxygenases (e.g., *KAO2, GA3, GA1*), which produce enzymes that catalyze the reactions from precursor molecules to several GAs, was greater in the gametophyte than in the sporophyte. Both GA_9 and GA_4 are vital for sex determination through the antheridiogen pathway in *Lygodium japonicum* (41) and upregulation of genes is expected as the gametophyte matures. Importantly, our samples contained a mixture of immature and mature gametophytes, which may be a confounding variable as the production of gametangia or gametes could influence gene expression. Despite this, our gametophytes were a similar size to those that upregulated *KAO* and *GA20OX* in *L. japonicum* (41); therefore the upregulation of these genes in our samples may represent gametophytes developing gametangia. There are few studies that have explicitly examined genes involved in sex regulation and gametangia development in fern gametophytes. While Atallah et al. (43) identified >1,000 differentially expressed genes in *C. richardii* gametophytes treated with antheridiogen, (44) found that *L. japonicum* gametophytes across

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different developmental stages had remarkably similar expression profiles. Although unintended, our design may have captured differentially expressed genes involved in gametangia development, a unique aspect of gametophyte biology.

Downregulated genes in the gametophyte were enriched for the flavonoid biosynthesis pathway (10.53 F.E., adjusted $P = 1.77 \times 10^{-3}$; *SI Appendix*, Fig. S7), including chalcone synthase (*CHS*) and flavonoid 3′-monooxygenase (*TT7*) genes. Secondary compounds, such as flavonoids, play an important role in defenses against herbivory in ferns (45). While chemical responses in the sporophyte have shed light on fern resistance to herbivory (3), the gametophyte has largely been ignored in this regard and further investigation is needed, especially as there are active biocontrol programs implemented to mitigate invasive populations of *L. microphyllum* (26) that could take advantage of herbivory on the gametophyte life stage.

We further examined two classes of genes known to be involved in directing plant development. Using a phylogenomic approach, we identified 46 homeobox genes in the *Lygodium* genome annotation (Fig. 3 and *SI Appendix*, Fig. S8 and Dataset S12). Interestingly, we identified a fern-specific clade of putative homeobox genes that fell sister to the major clade of HD-ZIP I genes, similar to the findings in ref. 46. A total of six homeobox genes had significantly different expression levels in the gametophyte and sporophyte. All had higher expression in the sporophyte than the gametophyte, except for a single gene (g3472, homolog of *PDF2*). In *Arabidopsis, PDF2* is essential for epidermal cell differentiation and downstream stomatal development (47); stomata are absent in fern gametophytes, suggesting that this gene may play another role in gametophyte development. In *Lygodium,* the downregulated genes in the gametophyte are members of the KNOX-II, BEL, and fern-specific HD-ZIP families, which appears to be a consistent pattern across millions of years of fern evolution, with significantly lower expression of KNOX-II genes in the gametophytes of *A. capillus-veneris* and *C. richardii* (Fig. 3*D*). KNOX genes are involved in sporophyte meristem development and leaf architecture in *Ceratopteris* (48), although few molecular developmental studies are available for ferns. While we cannot speculate on the functions of the fern-specific HD-ZIP genes, the lower expression levels of BEL and KNOX-II genes aligns with previous findings in bryophytes [e.g., *Anthoceros* (17)]. Specifically, KNOX-II genes in the sporophyte are required to maintain sporophyte cell fates in *P. patens* (49) and *M. polymorpha* (50). Our findings highlight a conservation of expression levels in these homeobox genes across >250 my of fern evolution, and perhaps >500 my of land plant evolution.

We also identified a total of 50 MADS-box genes in *Lygodium* (*SI Appendix*, Fig. S9 and Dataset S12). Of these, seven were significantly differentially expressed between sporophyte and gametophyte tissue; two genes (one in the CRM3 clade and one in MIKC* clade) were upregulated in the gametophyte, while the remaining were downregulated including three in the CRM6 clade (Fig. 3*C*). Interestingly, Zhang et al. (51) also identified several CRM3 genes with greater expression in the gametophyte than in the sporophyte, while CRM6-III genes exhibited the opposite pattern. There were clear differences in expression levels in CRM3 and CRM6 genes in *L. microphyllum* gametophyte and sporophyte tissues. Five MADS-box genes also had significantly different expression profiles in *Ceratopteris.* Four of these were downregulated in the gametophyte, including two CRM6 homologs, while an *AGL15* homolog was upregulated in mature gametophytes. In contrast to seed plants where expression of MADS-box genes have specific sporophytic and gametophytic expression patterns (52), MADS-box genes are widely expressed in both life phases in seed-free plants (7, 53), although the expression of genes such as CRM3 and CRM6 may play important roles in tissue differentiation between life stages. Both homeobox (54) and MADS-box genes (55) act as master regulators of plant development in both life phases; therefore, a comprehensive understanding of the regulation and expression of these genes in gametophyte and sporophyte development will undoubtedly shed light on the generation of morphologically and functionally different life phases from a single genome.

Posttranscriptional modifications, such as RNA splicing, are important processes that contribute to responses to environmental changes, gene regulation, and proteome diversity in plants (56). mRNA from a single gene in complex eukaryotes can be alternatively spliced into different transcripts, or isoforms, and can be classified into four main event types: exon skipping, intron retention, alternative donor, and alternative acceptor (Fig. 3*E*). Very little is known about splicing in ferns, although some authors have suggested it may play a role in development (e.g., ref. 57, but see ref. 8). We examined splicing events in the gametophyte and sporophyte to explore the posttranscriptional landscape in a homosporous fern. In alignment with data from other plants, the most common splicing event was intron retention (>40% of events, Fig. 3*E* and Dataset S13). Intron retention is an important mechanism in plant development and stress responses (58), and could have a role in promoting tissue-specific functions (59). Of the ~1,700 splicing events identified in each life phase, about a quarter were shared (404 total events); there were 1,363 and 1,302 unique events in the gametophyte and sporophyte, respectively (Fig. 3*E* and Dataset S13) and the proportion of each event was approximately equal when compared between tissue types. We compared which genes had unique splicing events in each life stage to the homeobox and MADS-box genes we identified above and manually verified the events. Of these events, the most interesting was an exon skipping event in the leaf for g22068, a fern-specific HD-ZIP gene. The removal of exon 2 in g22068 is a large portion of the leucine zipper including the DNA binding site, a strong indication that this isoform alters the function of the protein or renders it inactive (*SI Appendix*, Fig. S10). Whether alternative splicing is important in the context of gametophyte and sporophyte development in ferns requires further study, but our results reveal that there are patterns in both expression and splicing that are unique to each life stage.

Epigenomic Comparison of Fern Life Phases. DNA methylation is an important mechanism through which eukaryotes regulate gene expression without changing the underlying genetic code and may play a role in fern development (43, 60), but has only begun to be explored in seed-free plants (61). In plants, cytosine methylation is most common at CG sites across the genome, although it is also present in CHG and CHH (where H is A, T, or C) contexts (62). We reaffirmed that the genetic machinery for DNA methylation is largely present in fern genomes [*SI Appendix*, Figs. S11–S13; (6, 7)]. Using a phylogenomic approach, we identified homologs of genes involved in de novo methylation (*DRM1/2*), the maintenance of methylation (*DRM1/2, CMR1/2, DDM1,* and *MET1*), and demethylation (*ROS1*) that are present across eukaryotes (Dataset S14). It is possible that ferns also have homologs of the demethylation gene *DME*, although it was not possible for us to differentiate between *ROS1* and *DME* as their protein products have very similar motifs.

To explore the epigenomic landscape of ferns, we generated genome-wide methylation data at the single base resolution for a fern gametophyte (pooled immature and mature gametophytes) and compared these data to methylation patterns in the sterile leaf (hereafter sporophyte in this section). The genome of both life

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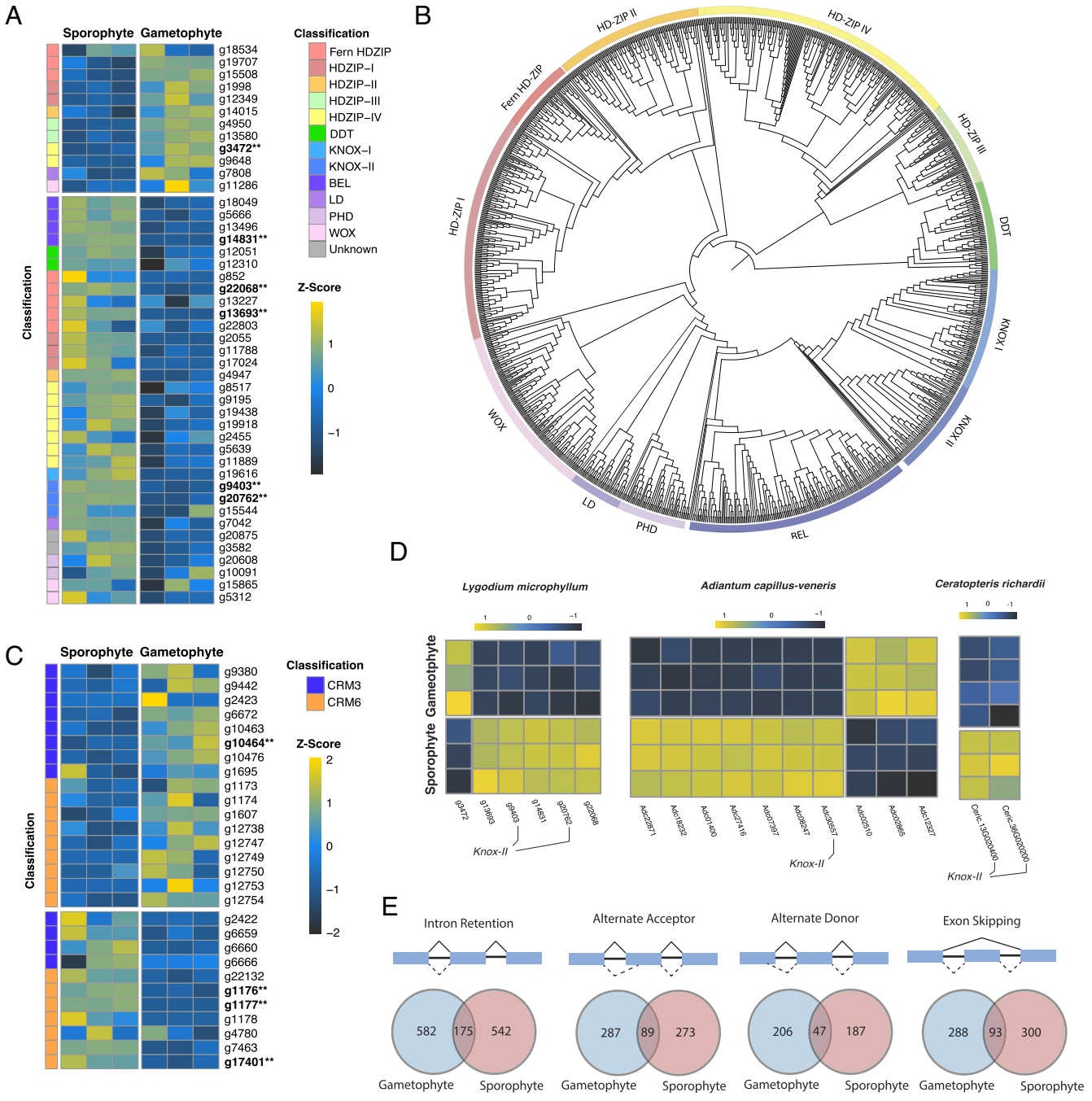


Fig. 3. Transcriptomic landscape of gametophyte and sporophyte life phases. (*A*) Heatmap of Z-scores of expression levels of homeobox genes identified in the *L. microphyllum* genome clustered based on expression level (blocks separated by horizontal gap). Significantly differentially expressed genes (log2FoldChange > 2, $P < 0.05$) are bolded and denoted with two asterisks (**). (*B*) Phylogeny of homeobox genes depicting the relationships of families across land plants. Classification colors are as in (*A*). (*C*) Heatmap of Z-scores of expression levels of CRM-3 and CRM-6 MADS-box genes identified in *Lygodium*. Significantly differentially expressed genes denoted with two asterisks (**) as in (*A*). (*D*) Significantly differentially expressed genes between sterile leaf and gametophyte tissue in three homosporous ferns. KNOX-II genes consistently have lower expression in the gametophytes of all three species. (*E*) The four classifications of alternative splicing events considered in this study, with numbers indicating unique events in the gametophyte (blue) and leaf (red) tissues, and those that are shared (intersections).

phases was highly methylated, although the gametophyte had lower methylation (mean 72.63% CpG sites methylated) compared to the sporophyte (93.75% CpG), similar moderate CHG methylation levels (62.70% in the gametophyte, 69.43% in the sporophyte), and low levels of CHH methylation (1.90% in the gametophyte, 1.46% in the sporophyte, Dataset S15). Both the CG and CHG methylation levels in *Lygodium* are similar to those reported for leaf methylation levels in the tree fern *A. spinulosa* (88.87% CpG, 66.83% CHG), although the CHH methylation level appears to be slightly greater in *Lygodium* (1.68%) compared to *Alsophila* (0.03%) (6). DNA methylation in plants is typically used to silence TEs (63), and given the high density of repeats in fern genomes it is unsurprising to see consistently high DNA methylation levels across the genome [Fig. 4; (6)]. Interestingly, very few sites were significantly differentially methylated between the gametophyte and sporophyte; 13,513 CpG sites were hypermethylated in the sporophyte, while just 7,229 CpG sites were hypomethylated in the sporophyte relative to levels in the gametophyte (Fig. 4). The majority of differentially methylated sites were located in intergenic regions (70.34% hypermethylated sites,

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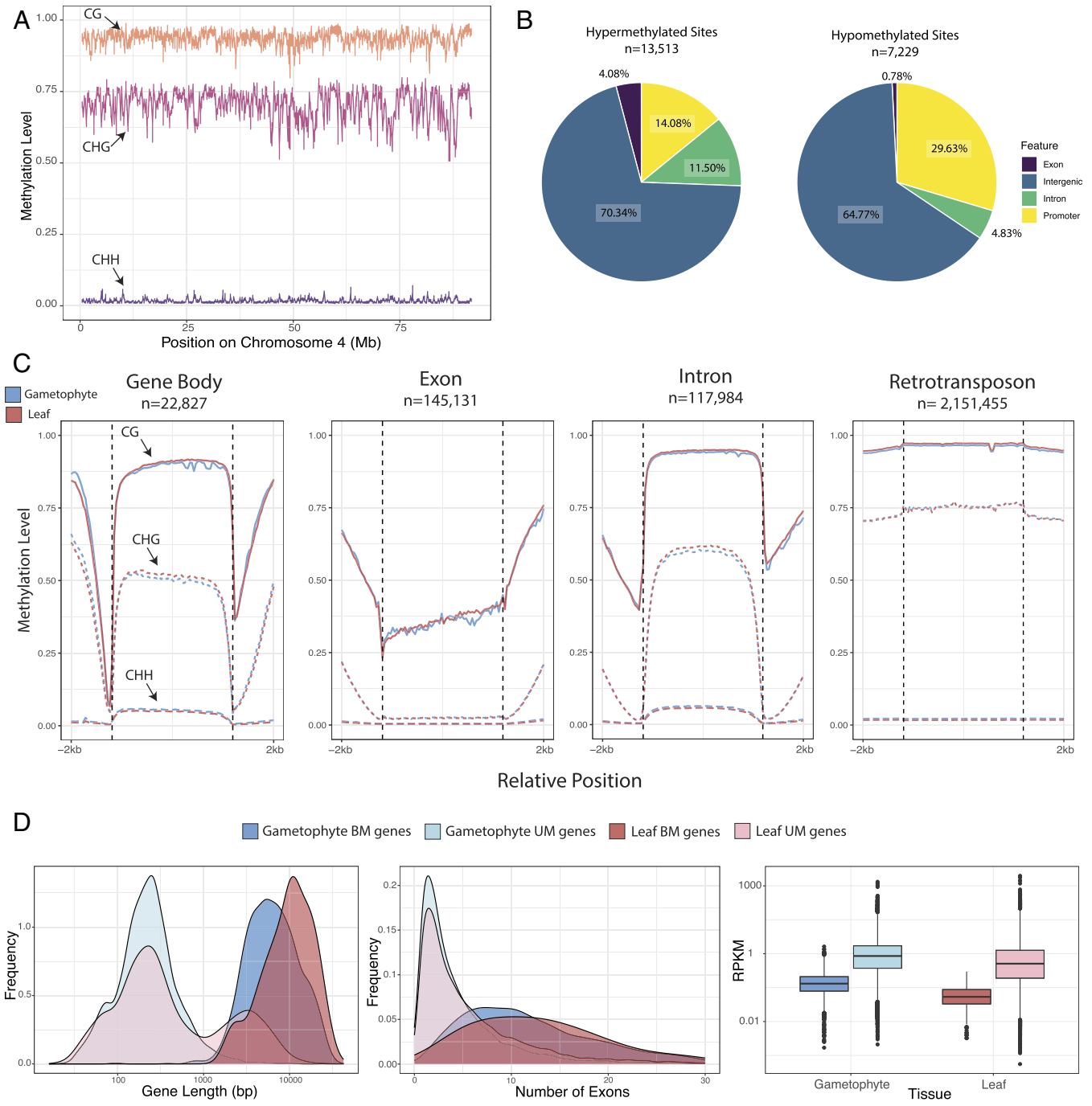


Fig. 4. Epigenomic differences in gametophyte and leaf tissues. (*A*) Distribution of DNA methylation in CG (orange), CHG (magenta), and CHH (purple) contexts over chromosome 4, as an example, in *Lygodium*. (*B*) Classification of regions with differentially methylated sites between gametophyte and leaf tissues. Hypomethylated sites are those that are significantly less methylated in the leaf than the gametophyte. Hypermethylated sites are those that are significantly more methylated in the leaf than the gametophyte. (*C*) Distribution of DNA methylation in CG (solid line), CHG (short dashed line), and CHH (long dashed line) contexts over the entire gene body, exons, introns, and retrotransposons for gametophyte (blue). (*D*) Plots showing the distribution of gene length, number of genes, and expression level (RPKM) for body-methylated (BM) and unmethylated (UM) genes identified in the gametophyte and leaf tissues.

64.77% hypomethylated sites), followed by putative promoters (14.08% hypermethylated sites, 29.63% hypomethylated sites). When we examined the putative functions of genes within 10 kb of these differentially methylated sites, the hypomethylated regions were associated with protein kinases (GO:0016773, GO:0004672, GO:0016301). No other functions were enriched in these regions. Sex determination in *Ceratopteris* gametophytes appears to be accompanied by large-scale epigenetic changes (43), although there do not appear to be widespread differences in the epigenome of the gametophytes and sporophytes in *Lygodium*.

Gene body methylation (gbM) is particularly interesting to plant biologists as its function is not fully understood and appears to be regulated through a different mechanism than the methylation of TEs (64). Researchers have proposed that gbM may be involved in the stabilization of gene expression, upregulation of gene expression, or preventing TE insertion into vital genes (64). Recent genomic studies have revealed that gbM is relatively common in ferns (6, 7, 65), but the gametophyte has not been included in these analyses thus far. To explore how gbM is distributed in the life phases of a fern, we compared gbM between gametophyte and

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sporophyte tissue in *Lygodium.* We identified substantially more body methylated (BM) genes in the gametophyte (2,033) compared to the leaf tissue (162), with the opposite trend for unmethylated (UM) genes (10,101 in the gametophyte, 16,125 in the leaf, Dataset S16). Nearly all the leaf BM genes also exhibited gbM in the gametophyte (144 shared BM genes), and most UM genes in the gametophyte were also UM in the leaf (9,915 shared UM genes). The proportion of gametophyte genes classified as BM (ca. 20%) is similar to that in *Arabidopsis* [ca. 27% of genes (66)] and across angiosperms [ca. 18% (67)], whereas the proportion of BM genes in the leaf is substantially lower (ca. 1%). Such a vast difference could be due to higher background methylation levels in the leaf tissues, making it difficult to pull out significant gbM using the binomial approach used here. Alternatively, differences in the number of BM genes could be due to extensive changes in DNA methylation in the life cycle, which are known to occur in *Marchantia polymorpha* (68) and *Arabidopsis thaliana* (69). gBM appears to be initiated by *CMT3* and its maintenance controlled by feedback between *CMT3* and histone H3K9me (70); therefore, given our methylation results, we may expect to see greater expression of *CMT3* in the gametophyte relative to the leaf.

Orthologs of *CMT3* have not been identified in fern genomes, and the molecular machinery through which gbM is established in ferns is unknown. Nonetheless, we explored the expression patterns of the methylation-related genes we could identify in *Lygodium*; nearly all (g2066, g18059, g4434, g3874, and g4450) had higher expression levels in the leaf, and only g22221 and g6614 had had greater expression values in the gametophyte (*SI Appendix*, Fig. S14). Interestingly, g22221 is an ortholog of *DDM1,* a chromatin remodeler that may play a role in gbM in *Arabidopsis* (71). Further study of the role of *DDM1* in ferns will be necessary to determine whether and how *DDM1* is associated with gbM.

To determine whether BM genes are enriched for functional categories, we performed a gene ontology (GO) enrichment test of the BM genes independently in each tissue relative to the total genome annotation. Housekeeping genes, which have constitutive and moderate expression, generally exhibit gbM in invertebrates (72). Our results largely agree with these findings (Dataset S17): both BM genes in the gametophyte and leaf were enriched for processes related to lipid metabolism (e.g., GO0042157, GO:0042158), while the gametophyte BM genes were also enriched for protein localization (GO0033365, GO:0008104) and hydrolase activity (e.g., GO:0016787, GO:0016811). Interestingly, in both gametophyte and leaf tissues BM genes had lower expression levels (measured by RPKM within samples) compared to UM genes (Fig. 4*D*), which contrasts with findings in *Arabidopsis* where BM genes were moderately expressed and UM genes had more variable expression profiles (66). The general characteristics of the BM genes, however, were similar to what is known in angiosperms. Consistent with the expectation that gbM is associated with stabilizing expression (66), gene length in BM genes was significantly greater than in UM genes in both the gametophyte (exact permutation test estimated by Monte Carlo with 999 permutations; $P = 0.002$) and leaf tissues ($P = 0.002$); the leaf BM genes were significantly longer than the gametophyte BM genes on average ($P = 0.002$; Fig. 4*D*). We also observed that BM genes had a greater number of exons in the gametophyte ($P = 0.002$) and leaf ($P = 0.002$), but there was not a difference between the gametophyte and leaf BM genes ($P = 0.396$; Fig. 4*D*).

Our analysis of the methylome of a fern gametophyte has revealed similar patterns of methylation in the haploid gametophyte life phase compared to the sporophyte. A slightly lower global methylation level in the gametophyte did not correlate with the very few differentially methylated sites between life phases. Our approach using pooled gametophyte tissues may swamp out the signal of tissue-specific methylation patterns in the gametangia and gametes that was observed in *Marchantia* (68). Epigenetic marks are an important mechanism through which plants can regulate gene expression and may play a vital role in the differentiation of gametophyte and sporophyte genetic programs, and further investigation will be required to untangle these processes in plants.

Phase-Specific Cold Tolerance Highlights Resilience of the Gametophyte. Freezing is widely predicted to be the most likely abiotic factor limiting range expansion of *L. microphyllum* (73). We used an ecological niche model to identify specific bioclimatic variables with high permutation importance in describing the distribution of this species, which were isothermality (bio3), temperature seasonality (bio4), and mean temperature of the warmest quarter (bio10, Dataset S18). Given the importance of temperature in the niche suitability of this species, *L. microphyllum* is likely to experience a northward range shift as the climate continues to change (Fig. 5), and the tolerance of the gametophyte and sporophyte life stages to these limiting variables will determine the spread of its invasion.

To understand how expression of the genome might shape physiological responses to limiting environmental conditions at each life stage, we measured physiological responses to cooling and freezing stress in both the gametophyte and sporophyte, and compared gene expression in these tissues. Several large-scale cellular changes are expected to occur during freezing stress. Most importantly, ice will form in the apoplast, disrupting the cell membrane, causing cell dehydration (74) and inducing osmotic shock (75). These changes directly impact the efficiency of the photosynthetic machinery, particularly photosystem II (76). Therefore, we first compared the maximum quantum yield of photosystem II (F_v/F_m) of sporophyte and gametophyte samples during the response to and recovery from prolonged cold stress at 2 °C (Datasets S19–S21). As a measure of chlorophyll fluorescence, F_v/F_m can be used as a proxy for plant stress, where values closer to zero represent more stressed plants. There was a significant impact of life stage and duration of cooling at 2 °C on the efficiency of photosystem II (linear regression, $F_{3,118} = 78.32$, $P = 6.42 \times 10^{-14}$, adjusted $R^2 = 0.66$). Sporophytes had, on average, greater F_v/F_m values than gametophytes during exposure to cold temperatures (0.23 higher F_v/F_m values, $P = 2.48 \times 10^{-10}$) and for every 10 h of cooling stress, F_v/F_m decreased by 0.03 independent of life stage. While there was a significant effect of duration of recovery on F_v/F_m (i.e., when plants were removed from 2 °C and placed at 25 °C; log–linear regression, $F_{4,99} = 34.74$, $P = 2.37 \times 10^{-9}$, adjusted $R^2 = 0.58$), there was not a significant difference in the recovery of sporophytes and gametophytes ($P = 0.26$; Fig. 5).

When exposed to short-term freezing stress (2 h at −20 °C), there was a significant impact of freezing and life stage on measures of F_v/F_m ($F_{3,66} = 15.74$, $P = 7.95 \times 10^{-8}$, adjusted $R^2 = 0.39$), including a significantly faster decrease in F_v/F_m by 0.05 per hour in sporophytes compared to gametophytes at −20 °C ($P = 0.04$). Fronds were unable to recover from 2 h of freezing stress ($F_{3,101} = 250.81$, $P < 2.2 \times 10^{-16}$, adjusted $R^2 = 0.88$). Importantly, it is well documented that while fronds of *Lygodium* may die back during freezing events, the rhizome is typically tolerant and will resprout (77). We observed a similar phenomenon, with new fronds emerging from the rhizome within weeks of the freezing shock stress. Gametophytes returned to prefreezing F_v/F_m values within 24 h, but secondary exposure to −20 °C temperature for 4 h resulted in death of the gametophytes. While Hutchinson and

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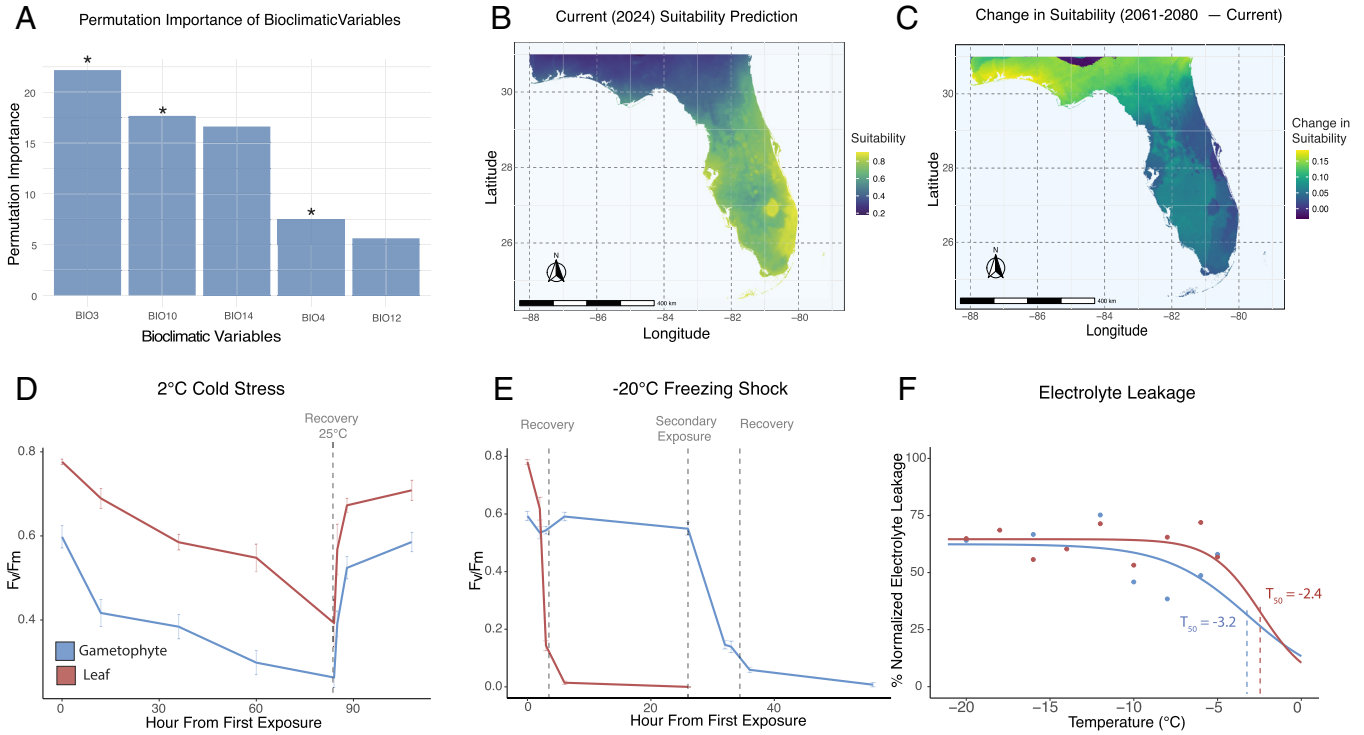


Fig. 5. Differential physiological responses to cold stress and freezing shock. (*A*) Permutation importance of climatic variables >5% within the niche model for *L. microphyllum*. Variables associated with temperature (bio3, bio10, and bio4) are denoted with asterisks. Predicted suitability of *L. microphyllum* during the present (*B*) and the difference in suitability when projected onto a future climate scenario for 2061 to 2080 and the present (*C*). A clear northward expansion in suitability is evident in the future climate scenario. (*D*) Gametophyte and leaf stress responses to exposure to 2 °C temperatures for 84 h and recovery at 25 °C for 24 h as measured by changes in the maximum quantum yield of photosystem II (F_v/F_m). (*E*) Responses of gametophyte and leaf to freezing shock (exposure to −20 °C) for 2 h and recovery at 25 °C for 24 h as measured by F_v/F_m. Gametophytes were exposed to a secondary, longer period (4 h) after recovery. Measurements of F_v/F_m were stopped once a plant had F_v/F_m values of 0 as these individual fronds and gametophytes never recovered from freezing stress. (*F*) Electrolyte leakage of gametophytes and leaf tissue, highlighting a slightly greater ability of the gametophyte to withstand freezing damage.

Langeland (27) observed a 7-fold decrease in survival of *L. microphyllum* gametophytes after just 15 min at −2.2 °C, we found that gametophytes were able to survive far longer durations of freezing stress. Gametophyte cell membranes were also able to withstand slightly lower freezing temperatures than fronds (gametophyte T_{50} = −3.2 °C, sporophyte T_{50} = −2.4 °C, Dataset S21), although a model including life stage did not improve model fit (ΔBIC = 11.5). Our results highlight that, while most management strategies have focused primarily on the sporophyte (e.g., biocontrol, herbicide), inhibiting the establishment of cold-tolerant gametophytes poses a unique management opportunity to prevent the further expansion northward in a changing climate. Targeting the gametophyte, in addition to ongoing mitigation of sporophyte populations, will improve management outcomes by capitalizing on differences in the susceptibility and resistance of each life phase to stressors employed in invasive species management.

Molecular Pathways Used in Response to Freezing Shock Are Tissue-Specific. Given the vastly different physiological responses of the sporophyte and gametophyte to freezing stress, we assessed transcriptomic changes associated with the leaf, gametophyte, and root tissues when exposed to freezing shock. The overall responses varied between tissue types: The leaf had a large number of differentially expressed genes in response to freezing (DEGs, 3903, 19.5% of the total), both the gametophyte (646, 3.3%) and root (415, 2.1%) had very few DEGs, suggesting that there is a massive transcriptomic change in the leaf but fewer changes in the gametophyte and root in response to freezing. We discuss the changes most striking pathways involved in responses to freezing below (Dataset S22 and *SI Appendix*, Figs. S15–S20).

The remobilization of starch has been implicated in successful abiotic stress responses in several plant species (78). Interestingly, the gametophyte had increased expression levels of genes in pathways enriched for metabolism, particularly in starch and sucrose metabolism (3.16 F.E., adjusted P = 0.0127; Fig. 6), while the same pathway was enriched in the downregulated genes of the leaf (1.96 F.E., adjusted P = 0.0454; Fig. 6). β-amylases, enzymes that act to release maltose that may act as an osmoprotectant under increased osmotic stress (79, 80), were enriched in these pathways and upregulated in the gametophyte but downregulated in the leaf. The gametophyte also showed enhanced expression leading to the accumulation of trehalose, a small sugar known to improve cold tolerance through several molecular mechanisms including increasing photosynthetic efficiency and ROS scavenging (81). Photosynthetic pathways (e.g., ath00195, 4.49 F.E., adjusted P = 0.0489) and proteins (e.g., ath00196, 18.84 F.E., adjusted $P = 3.12 \times 10^{-5}$) were also enriched in upregulated genes in the gametophyte, supporting the heightened efficiency of photosystem II observed during freezing stress in our physiological experiment (Figs. 5 and 6).

In contrast, the root tissue gene sets were not enriched for metabolic pathways (Fig. 6). Rather, upregulated genes were enriched for the carotenoid biosynthesis pathway (4.23 F.E., adjusted $P = 2.7 \times 10^{-5}$) and plant hormone signal transduction (11.91 F.E., adjusted P = 0.027; Fig. 6). Carotenoids, and molecules derived from them such as abscisic acid (ABA), are known to play important roles in abiotic stress responses in plants as free-radical scavengers and in regulating and directing plant architecture (82). Studies in angiosperms have highlighted heightened expression of carotenoid biosynthetic enzymes in roots under salt

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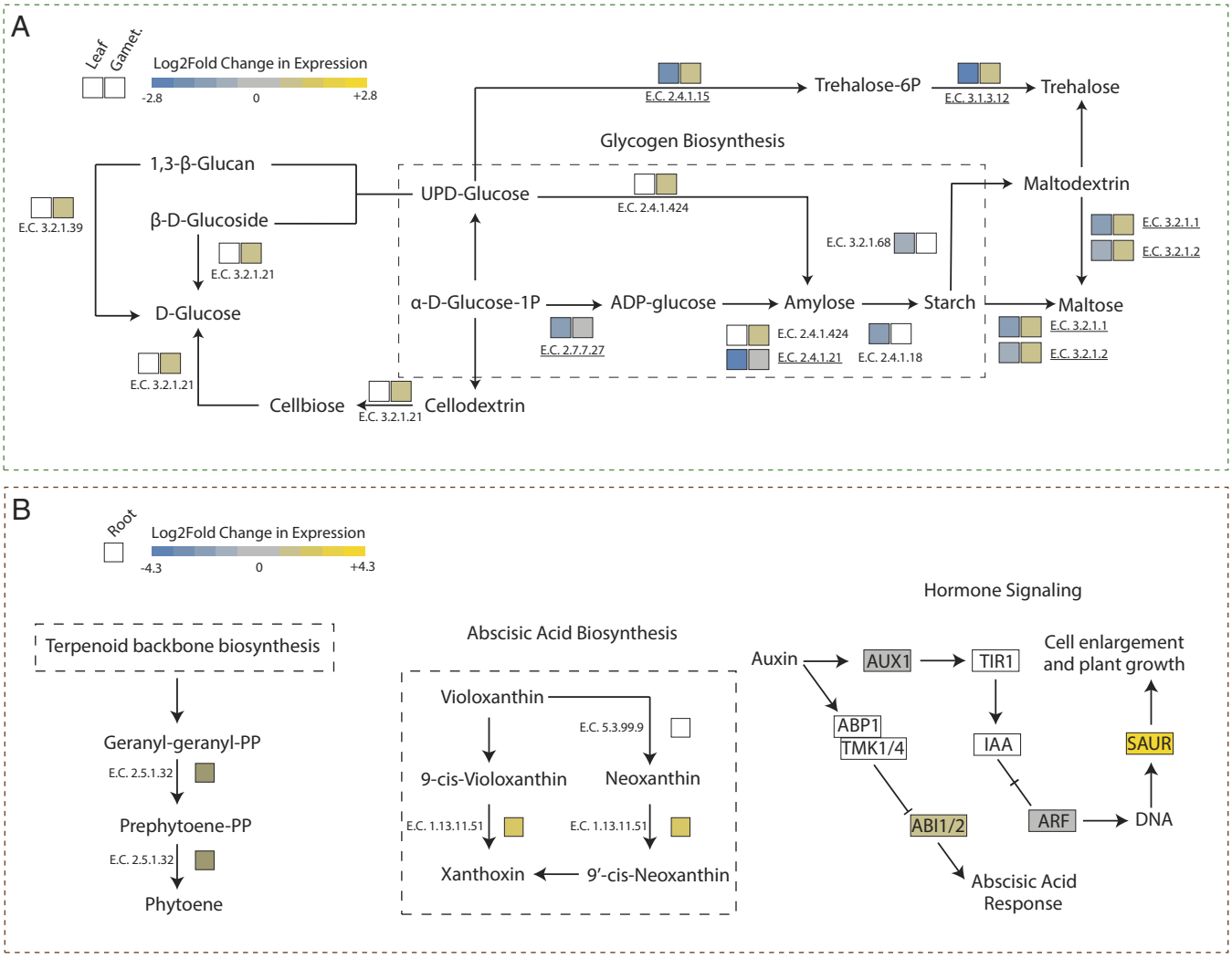


Fig. 6. Tissue-specific molecular pathways in response to freezing shock. (*A*) Subset of the starch and sucrose metabolism pathway, which was enriched in downregulated genes in the leaf and upregulated genes in the gametophyte as a response to freezing shock. Log2foldchange expression levels of genes involved in enzymatic reactions are listed at each reaction. (*B*) Subsets of the carotenoid biosynthesis and hormone signaling pathways that were enriched in upregulated genes in the root as a response to freezing shock. Log2foldchange expression levels of genes involved in enzymatic reactions or hormone-associated genes are listed at each reaction. White boxes are genes lacking expression data for that tissue type.

stress, but not shoots (83). Our results agree that the upregulation of carotenoid biosynthesis-related genes is tissue-specific and concentrated in root tissue.

Conclusion. In this study, we generated a chromosome-scale reference genome of the highly successful invasive vining fern *L. microphyllum* and explored the transcriptomic, epigenomic, and physiological differences between haploid gametophyte and diploid sporophyte life phases. We found that there are fine-scale transcriptomic differences in the regulation of developmental genes and splicing landscape that may explain, at least partially, how a single genome can produce two vastly different life stages. Using genome-wide methylation data for a fern gametophyte, we showed that there was remarkable similarity in the epigenomes of gametophytes and sporophytes. By pairing physiological metrics of stress and RNAseq, we were able to demonstrate that the different life stages of this fern likely leverage their shared genome via different strategies to respond to environmental changes. In the gametophyte, which showed heightened freezing tolerance, pathways involved in the accumulation of soluble sugars were activated, while the same pathways were downregulated in the leaf, which quickly succumbed to freezing shock. Our results highlight the tolerance of *L. microphyllum* gametophytes to freezing and suggest that the gametophyte may play a pivotal role in the establishment dynamics and range expansion of *L. microphyllum*, especially as the climate continues to warm. The success of invasive species has long intrigued biologists (84), and genomic sequencing in several plant systems has provided valuable insights into these processes (85, 86). The high-quality reference genome presented here will allow researchers to delve further into the genomics of colonization in a highly invasive fern. Given that many organisms exhibit functional variation between life stages, determining their underlying biology at multiple scales, from genome to phenotype and across life phases, is vital to developing a thorough understanding of organismal biology.

Materials and Methods

Living plant material (sterile leaf tissue) was collected in Taiwan from an individual currently maintained at the Dr. Cecilia Koo Botanic Conservation Center (voucher Lu 30898 available at the herbarium TAIF, collected from Mt. Tanfeng, Beitou District, Taipei City, Taiwan). For genome assembly, we used a combination of long-read Oxford Nanopore (~37×), Illumina short-read (~60×), and chromosome conformation capture (Dovetail's Omni-C; ~73×) sequencing technologies. A contig-level

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assembly was generated with error-corrected Nanopore reads >5 kb using HERRO v0.1.0 (87) and hifiasm v0.19.9 (88). The assembly was scaffolded with the Omni-C data using juicer v2.0 (89), YaHS v1.2 (90), and Juice Box Assembly Tools v2.17.00 (91). We used both reference protein and newly generated RNAseq data from nine tissue types to annotate the genome with the BRAKER3 v3.0.6 pipeline (92). Analysis of differential gene expression was performed with DESeq2 v1.44.0 (93) and alternative splicing events identified using a modified pipeline from Chamala et al. (94). MethylSeq data were mapped to the genome with bwa-meth v0.2.7 (https://github.com/brentp/bwa-meth) and analyzed with MethylDackel v0.6.1 (https://github.com/dpryan79/MethylDackel) and methylKit v1.18.0 (95). To compare gene body methylation (gbM) between gametophyte and sporophyte tissues, we used the probabilistic approach described by Takuno and Gaut (66). Details on the materials and methods used in this study can be found in *SI Appendix*.

Data, Materials, and Software Availability. Sequence data have been deposited in NCBI SRA under BioProject PRJNA1157352 (96) and from 10KP at https://db.cngb.org/search/organism/148566/ (97). The genome assembly is available at NCBI under Accession JBLQGR000000000 (98). Genome assembly and annotations are also available at FernBase (https://fernbase.org/) (99). All code used in this study is available at https://github.com/jessiepelosi/LygodiumGenome (100).

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1. C. H. Haufler *et al.*, Sex and the single gametophyte: Revising the homosporous vascular plant life cycle in light of contemporary research. *Bioscience* **66**, 928–937 (2016).
2. E. M. Sigel, E. Schuettpelz, K. M. Pryer, J. P. Der, Overlapping patterns of gene expression between gametophyte and sporophyte phases in the fern *Polypodium amorphum* (Polypodiales). *Front. Plant Sci.* **9**, 1450 (2018).
3. Y. Fang *et al.*, The genome of homosporous maidenhair fern sheds light on the euphyllophyte evolution and defenses. *Nat. Plants* **8**, 1024–1037 (2022).
4. Y. Geng *et al.*, A de novo transcriptome assembly of *Ceratopteris richardii* provides insights into the evolutionary dynamics of complex gene families in land plants. *Genome Biol. Evol.* **13**, evab042 (2021).
5. F.-W. Li *et al.*, Fern genomes elucidate land plant evolution and cyanobacterial symbioses. *Nat. Plants* **4**, 460–472 (2018).
6. X. Huang *et al.*, The flying spider-monkey tree fern genome provides insights into fern evolution and arborescence. *Nat. Plants* **8**, 500–512 (2022).
7. D. B. Marchant *et al.*, Dynamic genome evolution in a model fern. *Nat. Plants* **8**, 1038–1051 (2022).
8. N. Rahmatpour *et al.*, Analyses of *Marsilea vestita* genome and transcriptomes do not support widespread intron retention during spermatogenesis. *New Phytol.* **237**, 1490–1494 (2023).
9. E. J. Klekowski, H. G. Baker, Evolutionary significance of polyploidy in the Pteridophyta. *Science* **153**, 305–307 (1966).
10. C. N. Page, Ecological strategies in fern evolution: A neopteridological overview. *Rev. Palaeobot. Palynol.* **119**, 1–33 (2002).
11. E. S. Sorojsrisom, B. C. Haller, B. A. Ambrose, D. A. R. Eaton, Selection on the gametophyte: Modeling alternation of generations in plants. *Appl. Plant Sci.* **10**, e11472 (2022).
12. J. E. Watkins, M. Mack, T. R. Sinclair, S. S. Mulkey, Ecological and evolutionary consequences of desiccation tolerance in tropical fern gametophytes. *New Phytol.* **176**, 708–717 (2007).
13. J. Blake-Mahmud, E. B. Sessa, C. J. Visger, J. E. Watkins, Polyploidy and environmental stress response: A comparative study of fern gametophytes. *New Phytol.* **245**, 885–898 (2025).
14. T. Sato, A. Sakai, Cold tolerance of gametophytes and sporophytes of some cool temperate ferns native to Hokkaido. *Can. J. Bot.* **59**, 604–608 (1981).
15. T. Sato, Phenology and wintering capacity of sporophytes and gametophytes of ferns native to northern Japan. *Oecologia* **55**, 53–61 (1982).
16. A. Arun *et al.*, Convergent recruitment of TALE homeodomain life cycle regulators to direct sporophyte development in land plants and brown algae. *eLife* **8**, e43101 (2019).
17. F.-W. Li *et al.*, *Anthoceros* genomes illuminate the origin of land plants and the unique biology of hornworts. *Nat. Plants* **6**, 259–272 (2020).
18. T. Dierschke *et al.*, Gamete expression of TALE class HD genes activates the diploid sporophyte program in *Marchantia polymorpha*. *eLife* **10**, e57088 (2021).
19. B. Nelms, V. Walbot, Gametophyte genome activation occurs at pollen mitosis I in maize. *Science* **375**, 424–429 (2022).
20. M. Borg *et al.*, Epigenetic reprogramming rewires transcription during the alternation of generations in *Arabidopsis*. *eLife* **10**, e61894 (2021).
21. J. J. Smith *et al.*, The sea lamprey germline genome provides insights into programmed genome rearrangement and vertebrate evolution. *Nat. Genet.* **50**, 270–277 (2018).
22. P. Pyšek *et al.*, Scientists' warning on invasive alien species. *Biol. Rev. Camb. Philos. Soc.* **95**, 1511–1534 (2020).
23. C. Diagne *et al.*, High and rising economic costs of biological invasions worldwide. *Nature* **592**, 571–576 (2021).
24. Pteridophyte Phylogeny Group (PPG) I, A community-derived classification for extant lycophytes and ferns. *J. Syst. Evol.* **54**, 563–603 (2016).
25. R. W. Pemberton, A. P. Ferriter, Old World climbing fern (*Lygodium microphyllum*), a dangerous invasive weed in Florida. *Am. Fern J.* **88**, 165–175 (1998).
26. M. C. Smith *et al.*, Efforts to control *Lygodium microphyllum* in Florida and developments in biology and management. *CABI Rev.* **20**, 0040 (2025), 10.1079/cabireviews.2025.0040.
27. J. T. Hutchinson, K. A. Langeland, Tolerance of *Lygodium microphyllum* and *L. japonicum* spores and gametophytes to freezing temperature. *Invasive Plant Sci. Manag.* **7**, 328–335 (2014).
28. J. Goolsby, Potential distribution of the invasive old world climbing fern, *Lygodium microphyllum* in North and South America. *Nat. Areas J.* **24**, 351–353 (2004).
29. L.-Y. Kuo, F.-W. Li, A roadmap for fern genome sequencing. *Am. Fern J.* **109**, 212–223 (2019).
30. I. Manton, W. A. Sledge, Observations on the cytology and taxonomy of the pteridophyte flora of Ceylon. *Philos. Trans. R. Soc. B: Biol. Sci.* **238**, 127–185 (1954).
31. J. A. Pelosi *et al.*, *Lygodium japonicum* (Lygodiaceae) is represented by a tetraploid cytotype in Florida. *Am. Fern J.* **113**, 43–55 (2023).
32. J. A. Pelosi, E. H. Kim, W. B. Barbazuk, E. B. Sessa, Phylotranscriptomics illuminates the placement of whole genome duplications and gene retention in ferns. *Front. Plant Sci.* **13**, 882441 (2022).
33. One Thousand Plant Transcriptomes Initiative, One thousand plant transcriptomes and the phylogenomics of green plants. *Nature* **574**, 679–685 (2019).
34. C.-H. Huang, X. Qi, D. Chen, J. Qi, H. Ma, Recurrent genome duplication events likely contributed to both the ancient and recent rise of ferns. *J. Integr. Plant Biol.* **62**, 433–455 (2019), 10.1111/jipb.12877.
35. H. Chen *et al.*, Revisiting ancient polyploidy in leptosporangiate ferns. *New Phytol.* **237**, 1405–1417 (2023).
36. Z. Li, S. P. Kinosian, S. H. Zhan, M. S. Barker, Ancient polyploidy and low rate of chromosome loss explain the high chromosome numbers of homosporous ferns. bioRxiv [Preprint] (2024). https://doi.org/10.1101/2024.09.23.614530 (Accessed 1 January 2025).
37. C. H. Haufler, D. E. Soltis, Genetic evidence suggests that homosporous ferns with high chromosome numbers are diploid. *Proc. Natl. Acad. Sci. U.S.A.* **83**, 4389–4393 (1986).
38. J. A. Pelosi *et al.*, All tangled up: Unraveling phylogenetics and reticulate evolution in the vining ferns, *Lygodium* (Schizaeales). *Am. J. Bot.* **111**, e16389 (2024).
39. M. Kessler, D. Aros-Mualin, The power of independent generations in plants. *New Phytol.* **245**, 440–442 (2024), 10.1111/nph.20162.
40. D. E. Richards, K. E. King, T. Ait-Ali, N. P. Harberd, How gibberellin regulates plant growth and development: A molecular genetic analysis of gibberellin signaling. *Annu. Rev. Plant Physiol. Plant Mol. Biol.* **52**, 67–88 (2001).
41. J. Tanaka *et al.*, Antheridiogen determines sex in ferns via a spatiotemporally split gibberellin synthesis pathway. *Science* **346**, 469–473 (2014).
42. W. Dopp, Eine die antheridienbildung bei farnen fordernde substanz in den prothallien von *Pteridium aquilium* (L.) Kuhn. *Ber. Dtsch. Bot. Ges.* **63**, 139–147 (1950).
43. N. M. Atallah, O. Vitek, F. Gaiti, M. Tanurdzic, J. A. Banks, Sex determination in *Ceratopteris richardii* is accompanied by transcriptome changes that drive epigenetic reprogramming of the young gametophyte. *G3* **8**, 2205–2214 (2018).
44. K. Aya *et al.*, De novo transcriptome assembly of a fern, *Lygodium japonicum*, and a web resource database, Ljtrans DB. *Plant Cell Physiol.* **56**, e5 (2015).
45. A. Castrejón-Varela, B. Pérez-García, J. A. Guerrero-Analco, K. Mehltreter, A brief review of phytochemical defenses of ferns against herbivores. *Am. Fern J.* **112**, 233–250 (2022).
46. F. Romani, R. Reinheimer, S. N. Florent, J. L. Bowman, J. E. Moreno, Evolutionary history of homeodomain leucine zipper transcription factors during plant transition to land. *New Phytol.* **219**, 408–421 (2018).
47. K. U. Torii, Stomatal development in the context of epidermal tissues. *Ann. Bot.* **128**, 137–148 (2021).

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48. R. Sano *et al.*, Knox homeobox genes potentially have similar function in both diploid unicellular and multicellular meristems, but not in haploid meristems. *Evol. Dev.* **7**, 69–78 (2005).
49. K. Sakakibara *et al.*, KNOX2 genes regulate the haploid-to-diploid morphological transition in land plants. *Science* **339**, 1067–1070 (2013).
50. T. Dierschke *et al.*, Control of sporophyte secondary cell wall development in *Marchantia* by a class II KNOX gene. *Curr. Biol.* **34**, 5213–5222.e5 (2024), 10.1016/j.cub.2024.09.061.
51. R. Zhang *et al.*, Dynamic evolution of MADS-box genes in extant ferns via large-scale phylogenomic analysis. *Front. Plant Sci.* **15**, 1410554 (2024).
52. G. Theissen *et al.*, A short history of MADS-box genes in plants. *Plant Mol. Biol.* **42**, 115–149 (2000).
53. M. Hasebe, C. K. Wen, M. Kato, J. A. Banks, Characterization of MADS homeotic genes in the fern *Ceratopteris richardii*. *Proc. Natl. Acad. Sci. U.S.A.* **95**, 6222–6227 (1998).
54. R. L. Chan, G. M. Gago, C. M. Palena, D. H. Gonzalez, Homeoboxes in plant development. *Biochim. Biophys. Acta.* **1442**, 1–19 (1998).
55. C. Smaczniak, R. G. H. Immink, G. C. Angenent, K. Kaufmann, Developmental and evolutionary diversity of plant MADS-domain factors: Insights from recent studies. *Development* **139**, 3081–3098 (2012).
56. W. B. Barbazuk, Y. Fu, K. M. McGinnis, Genome-wide analyses of alternative splicing in plants: Opportunities and challenges. *Genome Res.* **18**, 1381–1392 (2008).
57. T. C. Boothby, R. S. Zipper, C. M. van der Weele, S. M. Wolniak, Removal of retained introns regulates translation in the rapidly developing gametophyte of *Marsilea vestita*. *Dev. Cell* **24**, 517–529 (2013).
58. J.J.-L. Wong, A. Y. M. Au, W. Ritchie, J. E. J. Rasko, Intron retention in mRNA: No longer nonsense: Known and putative roles of intron retention in normal and disease biology. *Bioessays* **38**, 41–49 (2016).
59. V. Rocchi, M. Janni, D. Bellincampi, T. Giardina, R. D'Ovidio, Intron retention regulates the expression of pectin methyl esterase inhibitor (PMEI) genes during wheat growth and development. *Plant Biol. (Stuttg.)* **14**, 365–373 (2012).
60. J. M. McGrath, E. Pichersky, Methylation of somatic and sperm DNA in the homosporous fern *Ceratopteris richardii*. *Plant Mol. Biol.* **35**, 1023–1027 (1997).
61. P. Szövényi, A. Gunadi, F.-W. Li, Charting the genomic landscape of seed-free plants. *Nat. Plants* **7**, 554–565 (2021).
62. R. Lister *et al.*, Highly integrated single-base resolution maps of the epigenome in *Arabidopsis*. *Cell* **133**, 523–536 (2008).
63. M. J. Sigman, R. K. Slotkin, The first rule of plant transposable element silencing: Location, location, location. *Plant Cell* **28**, 304–313 (2016).
64. A. M. Muyle, D. K. Seymour, Y. Lv, B. Huettel, B. S. Gaut, Gene body methylation in plants: Mechanisms, functions, and important implications for understanding evolutionary processes. *Genome Biol. Evol.* **14**, evac038 (2022).
65. S. Takuno, J.-H. Ran, B. S. Gaut, Evolutionary patterns of genic DNA methylation vary across land plants. *Nat. Plants* **2**, 15222 (2016).
66. S. Takuno, B. S. Gaut, Body-methylated genes in *Arabidopsis thaliana* are functionally important and evolve slowly. *Mol. Biol. Evol.* **29**, 219–227 (2012).
67. C. E. Niederhuth *et al.*, Widespread natural variation of DNA methylation within angiosperms. *Genome Biol.* **17**, 194 (2016).
68. M. W. Schmid *et al.*, Extensive epigenetic reprogramming during the life cycle of *Marchantia polymorpha*. *Genome Biol.* **19**, 9 (2018).
69. D. Bouyer *et al.*, DNA methylation dynamics during early plant life. *Genome Biol.* **18**, 179 (2017).
70. A. J. Bewick *et al.*, On the origin and evolutionary consequences of gene body DNA methylation. *Proc. Natl. Acad. Sci. U.S.A.* **113**, 9111–9116 (2016).
71. S. Lee *et al.*, DDM1-mediated gene body DNA methylation is associated with inducible activation of defense-related genes in *Arabidopsis*. *Genome Biol.* **24**, 106 (2023).
72. S. Sarda, J. Zeng, B. G. Hunt, S. V. Yi, The evolution of invertebrate gene body methylation. *Mol. Biol. Evol.* **29**, 1907–1916 (2012).
73. J. A. Goolsby, Potential distribution of the invasive Old World climbing fern, *Lygodium microphyllum* in North and South America. *Nat. Areas J.* **24**, 351–353 (2004).
74. R. Pearce, Plant freezing and damage. *Ann. Bot.* **87**, 417–424 (2001).
75. J. Krasensky, C. Jonak, Drought, salt, and temperature stress-induced metabolic rearrangements and regulatory networks. *J. Exp. Bot.* **63**, 1593–1608 (2012).
76. F. Kaplan, C. L. Guy, RNA interference of *Arabidopsis* beta-amylase8 prevents maltose accumulation upon cold shock and increases sensitivity of PSII photochemical efficiency to freezing stress. *Plant J.* **44**, 730–743 (2005).
77. R. W. Pemberton, Northward range expansion in Florida of the invasive fern *Lygodium microphyllum* (Lygodiaceae). *Sida* **20**, 1759–1761 (2003).
78. M. Thalmann, D. Santelia, Starch as a determinant of plant fitness under abiotic stress. *New Phytol.* **214**, 943–951 (2017).
79. F. Kaplan, C. L. Guy, Beta-amylase induction and the protective role of maltose during temperature shock. *Plant Physiol.* **135**, 1674–1684 (2004).
80. M. Thalmann *et al.*, Regulation of leaf starch degradation by abscisic acid is important for osmotic stress tolerance in plants. *Plant Cell* **28**, 1860–1878 (2016).
81. A. Raza *et al.*, Trehalose: A sugar molecule involved in temperature stress management in plants. *Crop J.* **12**, 1–16 (2023), 10.1016/j.cj.2023.09.010.
82. D. Ke *et al.*, Carotenoid-derived bioactive metabolites shape plant root architecture to adapt to the rhizospheric environments. *Front. Plant Sci.* **13**, 986414 (2022).
83. M. Á. Ruiz-Sola, V. Arbona, A. Gómez-Cadenas, M. Rodríguez-Concepción, A. Rodríguez-Villalón, A root specific induction of carotenoid biosynthesis contributes to ABA production upon salt stress in *Arabidopsis*. *PLoS ONE* **9**, e90765 (2014).
84. H. G. Baker, G. L. Stebbins, Eds., "The genetics of colonizing species" in *Proceedings of the First International Union of Biological Sciences Symposia on General Biology* (1965).
85. K. M. Dlugosch, S. R. Anderson, J. Braasch, F. A. Cang, H. D. Gillette, The devil is in the details: Genetic variation in introduced populations and its contributions to invasion. *Mol. Ecol.* **24**, 2095–2111 (2015).
86. A. McGaughran *et al.*, Genomic tools in biological invasions: Current state and future frontiers. *Genome Biol. Evol.* **16**, evad230 (2024).
87. D. Stanojevic, D. Lin, P. Florez De Sessions, M. Sikic, Telomere-to-telomere phased genome assembly using error-corrected Simplex nanopore reads. bioRxiv [Preprint] (2024). https://doi.org/10.1101/2024.05.18.594796 (Accessed 1 August 2024).
88. H. Cheng, G. T. Concepcion, X. Feng, H. Zhang, H. Li, Haplotype-resolved *de novo* assembly using phased assembly graphs with hifiasm. *Nat. Methods* **18**, 170–175 (2021).
89. N. C. Durand *et al.*, Juicer provides a one-click system for analyzing loop-resolution Hi-C experiments. *Cell Syst.* **3**, 95–98 (2016).
90. C. Zhou, S. A. McCarthy, R. Durbin, YaHS: Yet another Hi-C scaffolding tool. *Bioinformatics* **39**, btac808 (2023).
91. O. Dudchenko *et al.*, The Juicebox assembly tools module facilitates de novo assembly of mammalian genomes with chromosome-length scaffolds for under $1000. bioRxiv [Preprint] (2018). https://doi.org/10.1101/254797 (Accessed 1 September 2024).
92. L. Gabriel *et al.*, BRAKER3: Fully automated genome annotation using RNA-seq and protein evidence with GeneMark-ETP, AUGUSTUS, and TSEBRA. *Genome Res.* **34**, 769–777 (2024).
93. M. I. Love, W. Huber, S. Anders, Moderated estimation of fold change and dispersion for RNA-seq data with DESeq2. *Genome Biol.* **15**, 550 (2014).
94. S. Chamala, G. Feng, C. Chavarro, W. B. Barbazuk, Genome-wide identification of evolutionarily conserved alternative splicing events in flowering plants. *Front. Bioeng. Biotechnol.* **3**, 33 (2015).
95. A. Akalin *et al.*, MethylKit: A comprehensive R package for the analysis of genome-wide DNA methylation profiles. *Genome Biol.* **13**, R87 (2012).
96. J. A. Pelosi *et al.*, Lygodium microphyllum genome sequencing and assembly. National Center for Biotechnology Information. https://www.ncbi.nlm.nih.gov/bioproject?term=PRJNA1157352. Deposited 5 September 2024.
97. 10,000 Plant Genomes Project, L.-Y. Kuo, F.-W. Li, Lygodium microphyllum. CNGBdb. https://db.cngb.org/data_resources/organism/148566. Deposited 13 May 2024.
98. J. A. Pelosi *et al.*, JBLQGR000000000 Lygodium microphyllum genome assembly. National Center for Biotechnology Information (NCBI). Deposited 5 September 2024.
99. F.-W. Li. Lygodium microphyllum. FernBase. https://fernbase.org. Deposited 1 September 2025.
100. J. A. Pelosi, LygodiumGenome. GitHub. https://github.com/jessiepelosi/LygodiumGenome. Deposited 6 March 2025.

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